Pizza Sales Analysis

13 business questions answered with MySQL

Joins | Subqueries | Aggregations | Window Functions

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Q1. Total number of orders placed

```
SELECT COUNT(order_id) AS total_orders
FROM orders;
```

Result

total_orders
21350

Insight

The pizzeria handled 21,350 orders over the period in the dataset.

Q2. Total revenue from pizza sales

```
SELECT ROUND(SUM(od.quantity * p.price), 0)
       AS total_revenue
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id;
```

Result

total_revenue
817,860

Insight

Joining order_details with pizzas gives price x quantity per line; summing it gives total sales of about 817.9K.

Q3. Highest-priced pizza

```
SELECT pt.name, p.price
FROM pizza_types pt
JOIN pizzas p
  ON pt.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```

Result

name	price
The Greek Pizza	35.95

Insight

The Greek Pizza is the most expensive item on the menu at 35.95.

Q4. Most common pizza size ordered

```
SELECT p.size,
       COUNT(od.order_details_id) AS total_ordered
FROM order_details od
JOIN pizzas p ON od.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY total_ordered DESC;
```

Result

size	total_ordered
L	18526
M	15385
S	14137
XL	544
XXL	28

Insight

Large is the most ordered size (counted per order line). XL and XXL are very rare.

Q5. Top 5 most ordered pizza types

```
SELECT pt.name, SUM(od.quantity) AS total_ordered
FROM pizza_types pt
JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY total_ordered DESC
LIMIT 5;
```

Result

name	total_ordered
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Insight

The top 5 are very close: only 82 pizzas separate #1 from #5, so demand is spread evenly.

Q6. Total quantity ordered per category

```
SELECT pt.category, SUM(od.quantity) AS total_ordered
FROM pizza_types pt
JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od ON od.pizza_id = p.pizza_id
GROUP BY pt.category
ORDER BY total_ordered DESC;
```

Result

category	total_ordered
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Insight

Classic leads clearly with 14,888 pizzas ordered.

Q7. Orders by hour of the day

```
SELECT HOUR(order_time) AS hour,
       COUNT(order_id) AS orders
FROM orders
GROUP BY HOUR(order_time);
```

Result

hour	orders
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920

Insight

Orders peak at lunch (12-13h). Partial output shown; run the query to see all hours.

Q8. Pizza types per category

```
SELECT category, COUNT(name) AS pizza_count
FROM pizza_types
GROUP BY category;
```

Result

category	count
Chicken	6
Classic	8
Supreme	9
Veggie	9

Insight

The menu has 32 pizza types. Chicken has the fewest, yet it still sells nearly as much as the other categories.

Q9. Average pizzas ordered per day

```
SELECT ROUND(AVG(quantity), 0) AS avg_per_day
FROM (
  SELECT o.order_date, SUM(od.quantity) AS quantity
  FROM orders o
  JOIN order_details od ON o.order_id = od.order_id
  GROUP BY o.order_date
) AS order_quantity;
```

Result

avg_per_day
138

Insight

A subquery sums pizzas per date first, then AVG() gives about 138 pizzas per day.

Q10. Top 3 pizza types by revenue

```
SELECT pt.name, SUM(p.price * od.quantity) AS revenue
FROM pizza_types pt
JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

Result

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

Insight

All top 3 earners are chicken pizzas. Revenue ranking differs from quantity ranking because of price.

Q11. Revenue contribution of each category (%)

```
SELECT pt.category,
  ROUND(SUM(od.quantity * p.price) / (
    SELECT SUM(od2.quantity * p2.price)
    FROM order_details od2
    JOIN pizzas p2 ON od2.pizza_id = p2.pizza_id
  ) * 100, 2) AS revenue_pct
FROM pizza_types pt
JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id
JOIN order_details od ON od.pizza_id = p.pizza_id
GROUP BY pt.category
ORDER BY revenue_pct DESC;
```

Result

category	revenue_pct
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Insight

A scalar subquery gives total revenue as the denominator. All four categories contribute 24-27%, a balanced mix.

Q12. Cumulative revenue over time

```
SELECT order_date,
  SUM(revenue) OVER (ORDER BY order_date)
    AS cum_revenue
FROM (
  SELECT o.order_date,
         SUM(od.quantity * p.price) AS revenue
  FROM order_details od
  JOIN pizzas p ON od.pizza_id = p.pizza_id
  JOIN orders o ON o.order_id = od.order_id
  GROUP BY o.order_date
) AS sales;
```

Result

order_date	cum_revenue
2015-01-01	2713.85
2015-01-02	5903.05
2015-01-03	7501.60
2015-01-04	9678.45
2015-01-05	12250.40

Insight

A window function with ORDER BY creates a running total of daily revenue (first rows shown).

Q13. Top 3 pizzas by revenue in each category

```
SELECT name, revenue FROM (
  SELECT category, name, revenue,
    RANK() OVER (PARTITION BY category
                 ORDER BY revenue DESC) AS rnk
  FROM (
    SELECT pt.category, pt.name,
           SUM(od.quantity * p.price) AS revenue
    FROM pizza_types pt
    JOIN pizzas p ON pt.pizza_type_id = p.pizza_type_id
    JOIN order_details od ON od.pizza_id = p.pizza_id
    GROUP BY pt.category, pt.name) AS a
) AS b
WHERE rnk <= 3;
```

Result

name	revenue
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

Insight

PARTITION BY restarts the ranking inside each category. Partial output shown.

Key Takeaways

- 21,350 orders generated about 817.9K in revenue
- Large pizzas and the Classic category lead in volume
- Orders peak at lunchtime (12-13h)
- Chicken pizzas earn the top 3 revenue spots
- Revenue is evenly split across categories (24-27% each)

Tools: MySQL Workbench | Full code on GitHub